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Safe Driving Buddy (specification document)

General description of the project

Safe Driving Buddy is a program designed to help drivers learn about accident-prone areas before traveling to unfamiliar locations in the United States. The United States has a strong road-trip culture, and many people travel to unfamiliar places every year. However, drivers are often unaware of accident-prone areas, dangerous intersections, or roads with a high record of traffic crashes. Safe Driving Buddy is a web application designed to help drivers make informed decisions before and during their trips. Users can enter a city and state, and the application will display a map showing traffic accident locations and other relevant safety information.

I will use the NHTSA Crash API to obtain traffic crash data for locations in the United States. For the initial version, Safe Driving Buddy will use a simple command-line interface (CLI). The user will enter a city and state, and the program will retrieve and display basic traffic crash information in the terminal. This will allow me to focus on the main functionality before developing a web-based interface.

Safe Driving Buddy would use a web-based GUI. I plan to use Flask for the web application and a simple frontend where users can enter a city and state and view crash information on an interactive map.

Task Vignettes (User activity "flow")

Enter a Travel Location:

Users are planning to drive to a city they are not familiar with and want to learn about traffic crashes so they can make mental model for the drive. The user starts Safe Driving Buddy in the command line. The program asks the user to enter a city and state, such as Ames, Iowa. Once they enter the city and state, the program will search for available crash data.

Technical details:

- Input: City and state entered by the user.
- Version 1 uses a command-line interface (CLI) in Python.
- The program checks that the user provided the necessary location information.
- The city and state are passed to the next step for retrieving crash information.

Retrieve and Process Crash Data:

After the user enters a city and state, Safe Driving Buddy uses the location information to retrieve available traffic crash data from the NHTSA Crash API. The program processes the returned information and organizes the crash records so that the user can easily access the crash records.

Technical details:

- The program sends a request for crash data using the location information.
- Data is retrieved from the NHTSA Crash API.
- Python processes and organizes the returned data.

View Crash Information

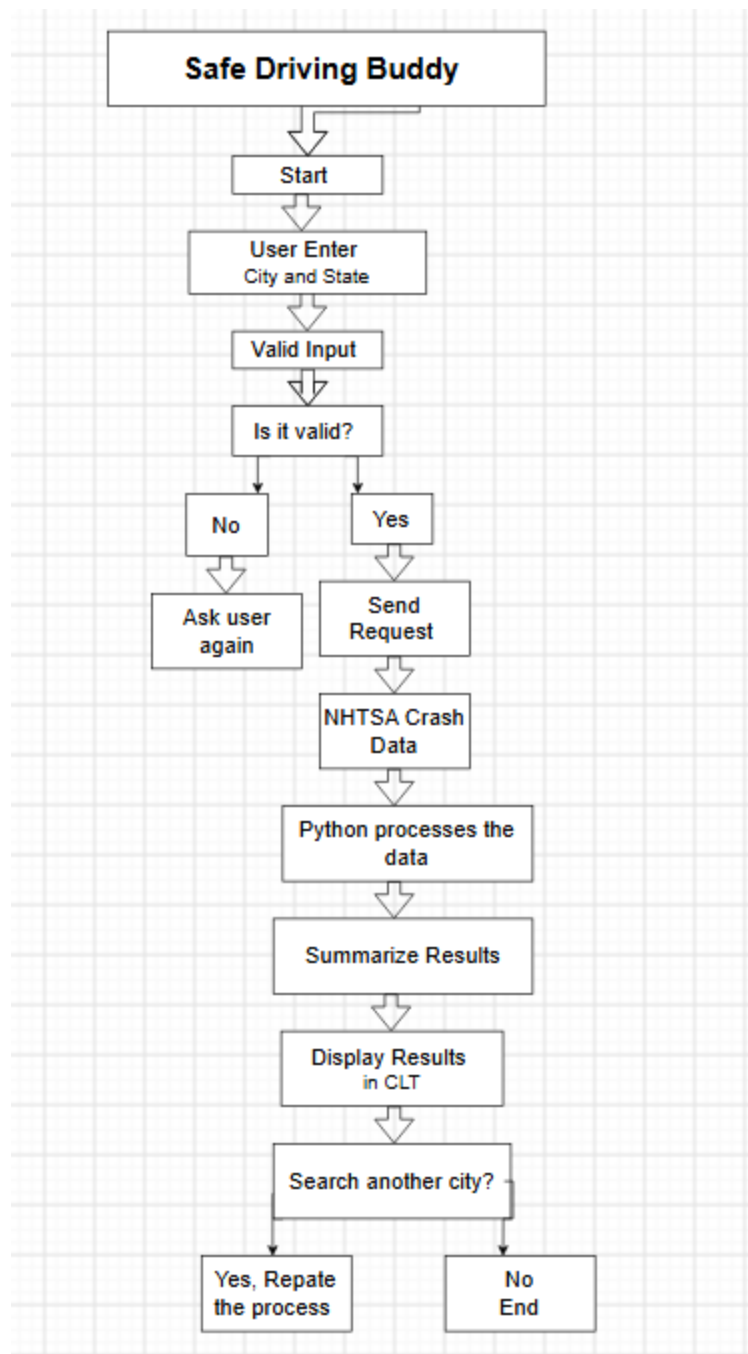
Once the crash information has been processed, Safe Driving Buddy displays the results in the terminal. The user might see information such as the number of crashes found and available crash-location information. The results give the user a better understanding of traffic safety concerns in the area before traveling.

Technical details:

- Output is displayed in the VS Code terminal for Version 1.
- The program summarizes the crash information in an easy-to-read format.
- Basic analysis could identify locations with multiple crashes.
- The user can search another city and state if desired.

Technical "flow"

Safe Driving Buddy will begin with a simple command-line interface (CLI). The user will enter a city and state. The Python program will use that information to retrieve available crash data and display basic results in the terminal. For the first version, I will focus on getting the input, retrieving the data, and displaying the output correctly. More advanced features, such as an interactive map and web interface, can be added later.



(Self) assesseemnt

The biggest change was simplifying the first version. Instead of starting with a web application and interactive map, I will begin with a simple CLI and add more features later.

I feel fairly confident that I can implement the project. Since I am still learning Python, breaking the project into smaller steps will make it more manageable.

The biggest challenge will be retrieving and processing the crash data. I need to make sure I can successfully access the data and display useful information to the user.

I may need help with accessing and processing the crash data in Python. Since working with an external API is new to me, I may also need guidance on choosing the appropriate data and keeping the first version of the project manageable.